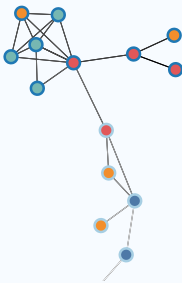


# SERGIO RUIZ-CARMONA

I work as a [Clinical Data Analyst](#), uncovering insights to drive informed decision-making. Proficient in extracting EMR records using SQL, I specialize in analyzing and visualizing data using R and Power BI. With a deep understanding of clinical data and a research background, I bring a unique perspective to optimizing healthcare outcomes.



View this resume [online](#)

## SELECTED EDUCATION

- Jan 2017  
|  
Jul 2012

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**PhD in Biomedicine**  
Universitat de Barcelona

Barcelona, ES

- Virtual Screening for novel MoA: Apps. and method developments
  - Development and application of **structure-based drug discovery** methods
  - Awards: **Extraordinary prize** for PhD Thesis and **Ramon Margalef award** for best publication
- Jun 2012  
|  
Sep 2010

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**MSc in Bioinformatics for Health Sciences**  
Universitat Pompeu Fabra

Barcelona, ES

- Expanding Druggable Genome: PP-Interfaces as a new target class
  - Study of novel rational approach to find PP interfacial binders
- Jun 2010  
|  
Sep 2006

●

**BSc in Biotechnology**  
Universitat Autònoma de Barcelona

Barcelona, ES

## WORK HISTORY

- Present  
|  
Mar 2025

●

**Data Analyst**  
Center for Child Health Analytics  
Luzerner Kantonsspital

Luzern, CH

- I turn clinical data into actionable insights to support evidence-based decisions. Experienced in extracting Epic EMR records using SQL and analyzing and visualizing data with R and Power BI, I help drive better healthcare outcomes.
- Feb 2025  
|  
Oct 2022

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**Clinical Data Analyst**  
Centre for Health Analytics  
Royal Children's Hospital

Melbourne, AUS
- Oct 2022  
|  
Jan 2019

●

**Senior Research Officer and Bioinformatician**  
Inouye Lab  
Baker Heart and Diabetes Institute

Melbourne, AUS

- I worked in the intersection between Structural Biology and Genomics, trying to understand how rare missense variants alter protein structures and can induce disease
- Dec 2019  
|  
Jan 2017

●

**Postdoctoral Researcher**  
Barril Lab, Universitat de Barcelona

Barcelona, ES

- After finishing my PhD, I worked in exciting collaborative projects in the field of epigenetics and cancer, where I carried out multidisciplinary research and learned new experimental skills

## SKILLS AND TRAINING

- **Computational Tools**  
Data visualization, Data analysis, Dashboards (Power BI, R Shiny), CI/CD, Machine learning, Statistical analysis, REDCap, Bioinformatics and Drug Discovery

## CONTACT

- ✉

[sruizcarmona@gmail.com](mailto:sruizcarmona@gmail.com)
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[sruizcarmona](#)
- in

[Sergio Ruiz](#)
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[ruizsergio.com](#)

## </> PROGRAMMING



## ⚡ HIGHLIGHTS

Highly experienced in:

- Computational biology and drug design
- Bioinformatics and biostatistics
- Machine Learning
- Data visualization
- Experimental techniques

Motivated and **collaborative**, recently involved in **Machine Learning**, NGS data analysis, **UK Biobank** and genomics prediction

## 📄 MORE INFO

See full CV at [cv.ruizsergio.com](http://cv.ruizsergio.com) for a complete list of positions, publications and more

Reference contact details provided upon request



Source code available in [github](#)  
Last updated in Aug, 2025

- **Programming**  
R, Python, SQL, Perl, C++, LaTeX, Bash, Java, HTML/CSS and Android and Web Development
- **Management Skills**  
2021 EMBO Practical Course: Research to service: Planning and running a bioinformatics core facility; Mental Health and OHS certifications
- **Certifications**  
Certified Health Informatician Australasia (CHIA), UX/UI Design (RMIT), Dataviz with R (IBM), Machine Learning (IBM)

## OUTREACH AND OTHER ACTIVITIES

- Sep 2021 | Sep 2020 ● **Australasian Leadership Computing Grants**  
Story about NCI Computing Grant
  - Together with Mike Inouye, we were awarded a 1 year computing grant by the Australian National Computational Infrastructure to study COVID-19 proteins and possible drug treatments
- Sep 2021 ● **New tools for new medicines, The Conversation [article](#)**  
Lay-summary of some of my work and opinion for The Conversation Spain
- Apr 2021 ● **[Shiny App](#) development**  
Related to a project we published in the Journal *Circulation*
- Dec 2020 ● **COVID-19 daily [dashboard](#), until Nov 2020**  
Personal project to showcase COVID-19 evolution in Victoria during major lockdown in 2020

## MAIN SCIENTIFIC OUTPUT

- Nov 2023 ● **COVID-19 thromboembolism is reduced in ambulatory, but not hospitalized patients, following COVID-19 vaccination**  
*Hematologica*  
Read it [here](#)
  - Hannah Stevens \*, **Sergio Ruiz-Carmona** \*, Karlheinz Peter and James D McFadyen
- Apr 2021 ● **Oxygen Pathway Limitations in Patients with Chronic Thromboembolic Pulmonary Hypertension**  
*Circulation*  
Read it [here](#)
  - Erin J Howden \*, **Sergio Ruiz-Carmona** \*, [...] Andre La Gerche, Marion Delcroix and Guido Claessen
  - Result of a Bioinformatics Core collaboration. [Shiny app](#) developed
- Jan 2017 ● **Dynamic undocking and the quasi-bound state as tools for drug discovery**  
*Nature Chemistry*  
Read it [here](#)
  - **Sergio Ruiz-Carmona**, Peter Schmidtke, [...] Rod Hubbard and Xavier Barril
  - Highlighted in its [issue cover](#)
- Sep 2014 ● **rDock: a fast, versatile and open source program for docking ligands to proteins and nucleic acids**  
*PLoS Computational Biology*  
Read it [here](#)
  - **Sergio Ruiz-Carmona**, Daniel Alvarez-Garcia, [...] Xavier Barril, Rod Hubbard and S David Morley

## EVERYDAY TOOLBOX

- RStudio
- Data viz & Shiny dashboards
- Bash, ssh & GitHub (local or HPC)
- ML optimization (genetic algorithms, MC, ...)
- Linear models & random forests

## WEB

I have also developed [rDock](#) and [Barril Lab](#) websites, used as dissemination platforms where the different tutorials and blog entries are shared with the community



I have been involved in more than 20 collaborative projects in very different fields. I enjoy helping other researchers and collaborators to work in common scientific challenges



My ultimate goal is to help people: from drug discovery to precision medicine and hospital care data analysis



Part of my research in drug discovery became a [drug patent](#) for cancer treatment, currently under development at [Oniria Therapeutics](#)